Sources of capital growth

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ABSTRACT

Data from national accounts support an argument that net saving, at collective scale, contributes nothing to growth in capital value. We explore ways in which this is possible, and discuss implications for economic teaching and public policy.

1. Introduction and overview

Many economists over the centuries have reasoned that net saving[1] should tend to give equal capital growth. Economists since the early nineteenth century have added the proviso that net saving cannot safely outpace innovation; more capital must mean capital redesigned for greater productivity if economies are to escape risk of capital glut and diminishing returns (West (1815), Ricardo (1815), Malthus (1815)). Roy Harrod (1939) described that limit for safe net saving, meaning the rate of imagining and developing new ideas for more productive forms of capital, as the "warranted rate". Harrod, and many other economists of his time and since, have focused on growth of output rather than of capital, but have modeled growth of output by first assuming the equivalence of net saving and capital growth, within the warranted rate, and then looking for effects of that capital growth on later output growth.

Some other economists, including John Rae (1834) and John Stuart Mill (1848) (see Appendix C), argued that capital growth might also be explained by a rise in productivity of capital and labor already extant. Ways might be found for existing factors to produce more, that is, and so to allow more consumption, or more capital growth, or any mix of the two, without inputs of net saving. Robert Solow (1957) allowed that possibility for "disembodied" growth, where plant and products already existing are repurposed or redeployed in more productive ways.

We test between those two explanations of capital growth, by net saving or by increase in productivity of capital and labor already in existence, by comparing net saving to concurrent change in market-value capital in 92 countries, and also comparing changes in those two variables. All data are drawn from national accounts of those countries as collated on the free website World Inequality Database. Our findings are easiest to reconcile in an interpretation that net saving, at the global scale, adds nothing to capital growth.[2] Net saving, that is, raises the physical quantity of capital, but not the aggregate value, and so reduces the value per unit. This condition describes capital glut.

The explanation we suggest is in production efficiencies of innovation. The innovator acquires materials and plant capacity and labor skills at market prices determined by their uses in current technology, but applies them more productively until competition catches up. It is that temporary market advantage to the innovator, which we call the innovator's reserve, that explains capital growth without net saving as a technical possibility.

The explanation in terms of source of funds is subtle. First, the innovator seeks potential investors just as with normal[3] rather than innovative business projects. These potential investors calculate present value, as with normal

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[1]As reported in national accounts; saving and investment differ only by statistical discrepancy. We view them as different, for reasons shown in Appendix A, but will equate them in this paper by using the word "saving" in the Keynesian sense of invested saving.

[2]We argue, however, that productivity gain will not give equal capital growth unless gross saving is enough to offset depreciation. We interpret any shortfall in this coverage as a negative component in net saving, and expect capital growth to equal productivity gain less that deficit (see Section 3).

[3]Let "normal" mean reliant on established rather than innovative technology.

projects, and invest up to that amount.[4,5] The subtle difference is that present value of normal projects tends to be just enough to meet initial costs of the project, while the innovator's reserve can bring present value above initial cost. We call this margin of present value above cost free growth, and call the mechanical and financial explanations together free growth theory.[6]

We further suggest, without evidence, that innovation should tend to bring positive externalities to capital of third parties not participating in the innovation. Old structures and infrastructure, for example, should tend to gain value as they house or support higher-tech workers and equipment. We call this effect "tech externalities." Free growth would then mean growth in capital of innovators through the innovator's reserve, and in capital generally through tech externalities.

Free growth theory predicts only at the largest scales, and only for the private sector. Individuals and groups and even small economies can grow in capital through saving if others dissave equally. That possibility is foreclosed only at the scale of all capital and all economies together. The public sector, meanwhile, responds to political rather than market choices, and grows or shrinks accordingly.

If free growth theory is right, tax policy and other policy to encourage saving over consumption should be reviewed. These policies include the higher tax on ordinary income than on capital gains, and the double tax on corporate dividends.

Inferences for economic teaching include the obvious ones for growth theory and for net saving in general. They include others as well. One of the central doctrines of the marginalist revolution has held that market realization converges to producer cost, when that cost includes imputed interest on assets owned. Net saving gives producer cost, and falls short of market realization in the presence of technological growth from new ideas. Meanwhile the doctrine that net income equals consumption plus net saving is put into question by evidence offered here suggesting that net saving increases the physical quantity of capital, but not the aggregate value.

2. Net saving, capital growth and capital acceleration

Thrift theory will mean the combined ideas that net saving S_{net} is realized in equal growth of capital K, if S_{net} holds within the warranted rate, and that there is no other source of capital growth except market noise regressing to zero. "Thrift assumptions" will mean those two plus the assumptions that current S_{net} holds within the warranted rate. The acronym uta will mean "under thrift assumptions." Then

$$\Delta K = S_{net} \,, \quad \text{uta.} \tag{1}$$

Define "thrift" s^* by $s^* = \frac{S_{net}}{K}$, and divide Eq. (1) by capital to predict capital growth rate $g(K)$ as

$$g(K) = s^* \,, \quad \text{uta,} \quad \text{where } s^* = \frac{S_{net}}{K}. \quad \text{Then} \tag{2}$$

$$\Delta g(K) = \Delta s^* \,, \quad \text{uta.} \tag{3}$$

$\Delta g(K)$ and Δs^* respectively will be called "capital acceleration" and "thrift change," either of which may be positive or negative or zero. Division of Eq. (3) by capital acceleration, with rearrangement, gives

$$\frac{\Delta s^*}{\Delta g(K)} = \frac{\Delta g(K)}{\Delta g(K)} = 1 \,, \quad \text{uta.} \tag{4}$$

Algorithms for correlation and covariance and regression are structured on the reasoning that causal relationships among variables tend to be revealed in concurrent changes. While Eqs. (1) and (2) enable measurement of the coexistence of net saving and capital growth, Eqs. (3) and (4) allow a better measure of the degree to which that coexistence is explained by causality. Eq. (4) was derived from Eq. (3) so that success of predictions from thrift

[4]Present value is the sum of time-discounted cash flows expected. These flows will be negative over any period of investment, and then positive as products are finished and sold.

[5]Competition among investors should tend to keep investment close to present value.

[6]The innovator may seek further funds at later stages. At each successive stage, present value is expected to have risen as positive cash flows approach with time. Investment is again expected to approximate present value, and again the innovator's reserve should tend to make the innovator's costs less than this amount raised. In each stage, that advantage to the innovator adds to free growth.

assumptions can be measured against a standard of unity. For notational convenience, then, we define the "thrift index" θ by $\theta = \frac{\Delta s^*}{\Delta g(K)}$, and restate Eq. (4) as

$$\theta = 1\,, \quad \text{uta,} \quad \text{where } \theta = \frac{\Delta s^*}{\Delta g(K)}\,. \tag{5}$$

Test results for the predictions $\frac{s^*}{g(K)} = \theta = 1$, under thrift assumptions, appear in Fig. 1 and in Tables 1, 2 and 3. Screens to which they refer are explained in Section 16 below. Results were found as GDP-weighted averages over all countries and years, from data for net saving and market-value capital taken from national accounts. Figures 1 and 2 show $\frac{s^*}{g(K)}$ and $\frac{\Delta s^*}{\Delta g(K)}$ as 0.481 and 0.064 respectively. Table 1 shows regression of s^* on $g(K)$ as 0.0771, and regression of Δs^* on $\Delta g(K)$ as 0.0559.

3. Interpretation of test results

The finding $\frac{s^*}{g(K)} = 0.481$ reveals average observed proportions between s^* and $g(K)$, we repeat, and does not of itself show causal relatedness between those two variables. Causal relatedness is shown rather by the degree of constancy in proportions, and thus on regression of s^* on $g(K)$ found at 0.0771.

θ or $\frac{\Delta s^*}{\Delta g(K)}$, measured at 0.064, gives a separate measure of causal relatedness between s^* and $g(K)$ as explained. Regression of Δs^* on $\Delta g(K)$, found at 0.0559, gives a measure of causal relatedness between Δs^* and $\Delta g(K)$. Might these small but positive findings allow the possibility that some capital growth is explained by net saving?

Free growth theory predicts that the present value of new ideas can pay for capital growth, but only after gross saving has made up for all depreciation. In years of recession, households may defer part or all of depreciation saving[7] to support consumption. In these years, growth in the present value of better ideas will not realize equal net growth of capital, as some of it must offset any shortfall in depreciation saving first. Allowing for these years, we can say more compactly that free growth theory predicts that a gain in the present value of new ideas accounts for all capital creation, and thus for the sum of capital growth and restoration of shortfalls in depreciation saving. Thus free growth theory does not predict a zero value for θ, but rather predicts θ to equal the share of capital creation applied to restoring those shortfalls. With this understanding, free growth theory could fit the appearance in Fig. 2 that θ found higher values during the economic reverses of the Iraq-Kuwait, dot com and subprime crises. We will suppose that 0.064 gives a plausible estimate of the shortfall in depreciation saving, until more is known, and will proceed on the simplifying assumption that free growth theory predicts correctly.

Table 1
Regression of s^* on $g(K)$ and Δs^* on $\Delta g(K)$, GDP-weighted. Screen = 0.01. H_0 per thrift theory: $\frac{s^*}{g(K)} \cong \frac{\Delta s^*}{\Delta g(K)} \cong 1$.

Regression of s^* on $g(K)$	0.0771***	Regression of Δs^* on $\Delta g(K)$	0.0559***
	(0.0144)		(0.0057)
Observations	1,826	Observations	1,574
R^2	0.84511	R^2	0.37071
Within R^2	0.10669	Within R^2	0.27233
Year fixed effects	✓	Year fixed effects	✓
Country fixed effects	✓	Country fixed effects	✓

[7] Also called replacement saving.

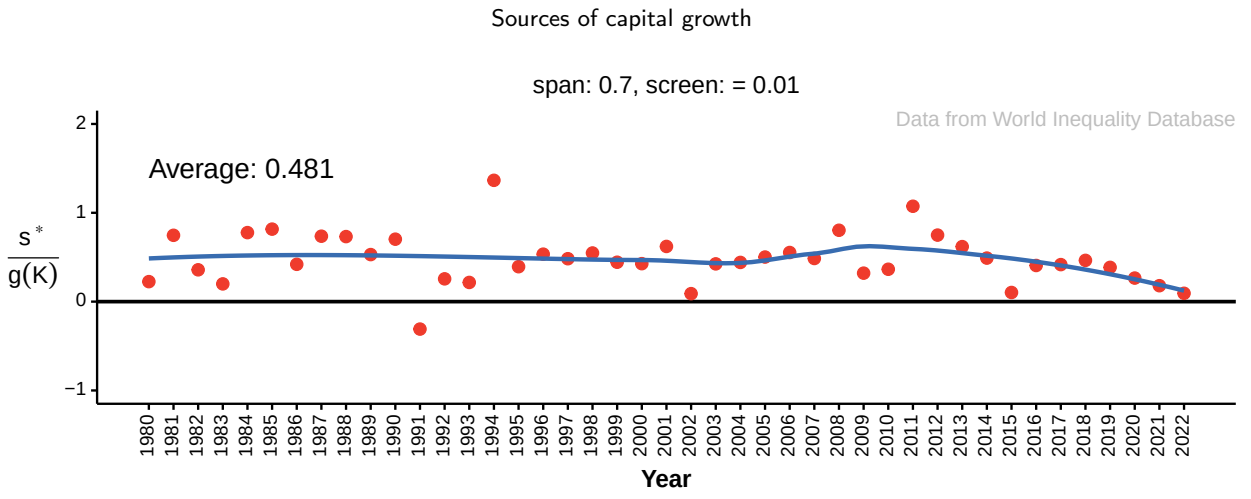


Fig. 1: Average $\frac{s^*}{g(K)}$ with LOESS smoothing over all countries, GDP-weighted, 1980-2022

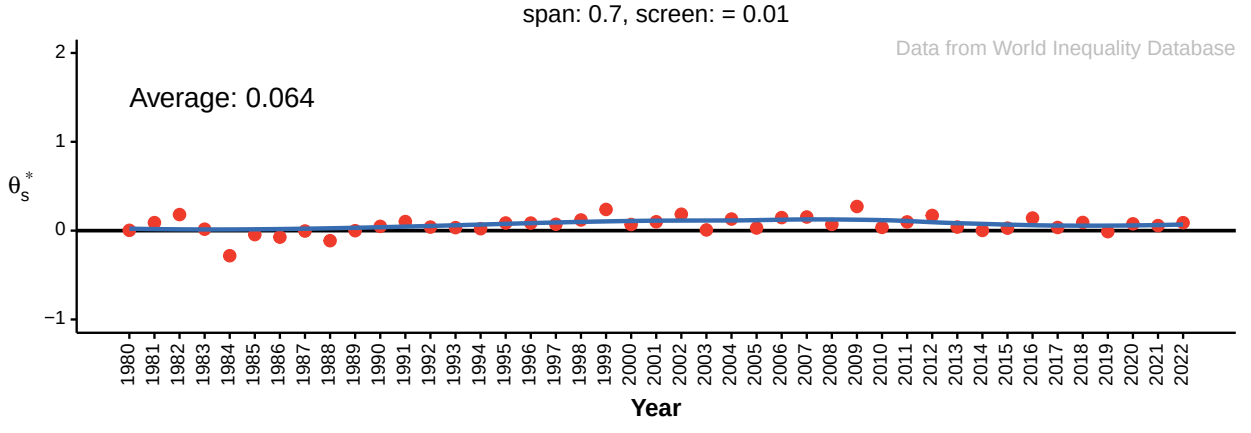


Fig. 2: Average θ_s^* with LOESS smoothing over all countries, GDP-weighted, 1980-2022

Table 2
Average $\frac{s^*}{g(K)}$ in 92 countries (screen = 0.01). Number of years clearing screen shown in ()

Armenia	1996 - 2020 (23)	-0.63	Italy	1980 - 2022 (36)	0.13
Aruba	1996 - 2001 (6)	1.25	Japan	1980 - 2021 (29)	0.19
Australia	1980 - 2018 (33)	0.24	Kazakhstan	1997 - 2022 (25)	0.60
Austria	1996 - 2021 (23)	0.83	Kuwait	2003 - 2017 (14)	1.11
Azerbaijan	1996 - 2020 (23)	1.57	Kyrgyzstan	1996 - 2021 (24)	0.46
Bahrain	2009 - 2013 (5)	-2.43	Latvia	1996 - 2021 (24)	-0.23
Belgium	1996 - 2021 (21)	0.61	Lithuania	1996 - 2021 (24)	0.13
Bolivia	1998 - 2015 (17)	0.40	Luxembourg	1996 - 2021 (22)	0.03
Botswana	1996 - 1999 (4)	3.54	Malaysia	2007 - 2015 (9)	2.52
Brazil	1996 - 2019 (21)	0.25	Malta	1996 - 2021 (23)	0.16
British Virgin Islands	1996 - 1999 (4)	2.45	Mauritius	2014 - 2018 (5)	0.04
Bulgaria	1996 - 2016 (16)	0.14	Mexico	1996 - 2021 (22)	0.15
Burkina Faso	2000 - 2019 (20)	0.23	Moldova	1996 - 2019 (22)	-0.91
Cameroon	1996 - 2019 (24)	1.10	Mongolia	2006 - 2020 (15)	0.10
Canada	1980 - 2022 (38)	0.37	Morocco	1999 - 2021 (23)	1.13
Cape Verde	2008 - 2017 (9)	0.53	Netherlands	1996 - 2021 (23)	0.51
Chile	1997 - 2021 (19)	0.84	New Zealand	1996 - 2019 (21)	0.78
China	1992 - 2020 (29)	0.81	Nicaragua	2007 - 2018 (12)	0.08
Colombia	1996 - 2022 (27)	0.60	Niger	1996 - 2019 (24)	0.90
Costa Rica	2013 - 2020 (6)	0.21	Norway	1981 - 2021 (38)	0.95
Croatia	1996 - 2021 (24)	0.43	Peru	2008 - 2021 (14)	0.76
Curaçao	2001 - 2016 (14)	1.26	Philippines	1996 - 2022 (27)	1.76
Cyprus	1996 - 2021 (25)	0.07	Poland	1996 - 2021 (25)	0.71
Czechia	1994 - 2020 (20)	0.17	Portugal	1996 - 2022 (24)	0.00
Côte d'Ivoire	1996 - 2000 (5)	0.29	Qatar	2002 - 2017 (14)	1.89
Denmark	1996 - 2022 (26)	0.22	Romania	1996 - 2020 (22)	0.03
Dominican Republic	1996 - 2016 (10)	1.52	Russia	1996 - 2017 (13)	-0.19
Ecuador	2008 - 2020 (13)	0.53	Saudi Arabia	2003 - 2009 (7)	2.65
Egypt	1997 - 2015 (19)	0.92	Senegal	2015 - 2021 (7)	0.24
El Salvador	2015 - 2019 (5)	-0.62	Serbia	1998 - 2021 (23)	0.61
Estonia	1996 - 2021 (24)	0.45	Slovakia	1996 - 2022 (24)	0.15
Finland	1996 - 2021 (23)	0.17	Slovenia	1996 - 2021 (20)	0.28
France	1980 - 2021 (35)	0.26	South Africa	1996 - 2022 (24)	0.28
Germany	1980 - 2022 (38)	0.70	South Korea	1996 - 2020 (24)	0.63
Greece	1996 - 2021 (26)	0.17	Spain	1995 - 2021 (25)	0.32
Guatemala	2002 - 2021 (20)	-0.72	Sweden	1980 - 2021 (37)	0.41
Guinea	2004 - 2010 (5)	0.72	Switzerland	1993 - 2021 (25)	0.69
Honduras	2001 - 2015 (14)	-0.57	Tunisia	1996 - 2011 (16)	0.29
Hong Kong	1997 - 2011 (15)	0.90	Turkey	2010 - 2017 (8)	1.32
Hungary	1996 - 2021 (21)	0.14	USA	1980 - 2021 (38)	0.31
Iceland	2001 - 2014 (14)	0.22	Ukraine	1996 - 2019 (21)	0.17
India	1999 - 2019 (21)	0.57	United Kingdom	1981 - 2021 (36)	0.16
Indonesia	2017 - 2019 (3)	0.94	Uruguay	2016 - 2017 (2)	1.16
Iran	1997 - 2017 (20)	4.96	Uzbekistan	2011 - 2021 (10)	0.21
Ireland	1996 - 2021 (22)	0.26	Vanuatu	2002 - 2007 (6)	0.29
Israel	1996 - 2019 (24)	0.46	Venezuela	1998 - 2019 (20)	0.55

Note: s^* is defined as $\frac{S_{net}}{K}$. Thrift theory predicts $\frac{s^*}{g(K)} \cong 1$. Free growth theory makes no prediction for these data.

Table 3
Average θ in 92 countries (screen = 0.01). Number of years clearing screen shown in ()

Armenia	1997 - 2020 (20)	0.12	Italy	1972 - 2021 (29)	0.05
Aruba	1997 - 2001 (5)	1.22	Japan	1972 - 2021 (38)	0.05
Australia	1974 - 2019 (35)	0.03	Kazakhstan	1997 - 2022 (23)	0.08
Austria	1997 - 2021 (16)	0.10	Kuwait	2005 - 2017 (11)	0.04
Azerbaijan	1997 - 2020 (23)	0.75	Kyrgyzstan	1998 - 2021 (22)	0.23
Bahrain	2010 - 2013 (4)	0.41	Latvia	1998 - 2021 (20)	0.01
Belgium	1997 - 2021 (19)	0.05	Lithuania	1997 - 2021 (18)	0.10
Bolivia	1999 - 2014 (12)	0.04	Luxembourg	1997 - 2021 (25)	0.00
Botswana	1997 - 2000 (4)	0.24	Malaysia	2008 - 2015 (7)	0.01
Brazil	1998 - 2020 (22)	0.10	Malta	1997 - 2021 (23)	0.08
British Virgin Islands	1997 - 1998 (2)	2.24	Mauritius	2015 - 2018 (4)	0.02
Bulgaria	1997 - 2017 (15)	-0.03	Mexico	1997 - 2021 (22)	0.06
Burkina Faso	2001 - 2019 (16)	0.04	Moldova	1997 - 2019 (21)	0.11
Cameroon	1997 - 2019 (19)	0.18	Mongolia	2008 - 2021 (11)	0.10
Canada	1974 - 2022 (41)	0.09	Morocco	2000 - 2021 (17)	0.19
Cape Verde	2009 - 2016 (8)	0.27	Netherlands	1997 - 2021 (20)	0.03
Chile	1998 - 2021 (22)	0.06	New Zealand	1997 - 2019 (18)	0.14
China	1981 - 2020 (33)	0.07	Nicaragua	2008 - 2019 (12)	0.06
Colombia	1997 - 2022 (18)	-0.03	Niger	1997 - 2019 (19)	0.22
Costa Rica	2014 - 2020 (6)	0.04	Norway	1982 - 2021 (34)	0.12
Croatia	1997 - 2021 (22)	0.19	Peru	2009 - 2021 (11)	0.04
Curaçao	2002 - 2016 (14)	0.41	Philippines	1997 - 2022 (20)	0.37
Cyprus	1997 - 2021 (23)	-0.06	Poland	1997 - 2021 (15)	0.20
Czechia	1995 - 2021 (19)	-0.03	Portugal	1997 - 2022 (17)	0.02
Côte d'Ivoire	1997 - 2000 (4)	-0.03	Qatar	2004 - 2018 (12)	0.29
Denmark	1997 - 2022 (22)	0.05	Romania	1997 - 2020 (14)	0.02
Dominican Republic	2007 - 2015 (9)	0.15	Russia	1997 - 2018 (14)	0.09
Ecuador	2009 - 2020 (12)	0.17	Saudi Arabia	2005 - 2009 (3)	2.23
Egypt	1998 - 2015 (16)	0.13	Senegal	2018 - 2021 (3)	0.19
El Salvador	2016 - 2019 (4)	0.98	Serbia	1999 - 2021 (22)	0.22
Estonia	1997 - 2021 (19)	0.06	Slovakia	1997 - 2021 (18)	0.09
Finland	1997 - 2022 (21)	-0.02	Slovenia	1997 - 2021 (20)	0.10
France	1972 - 2021 (33)	0.09	South Africa	1997 - 2022 (21)	0.10
Germany	1972 - 2022 (31)	0.05	South Korea	1997 - 2020 (15)	0.04
Greece	1997 - 2021 (24)	0.24	Spain	1972 - 2021 (42)	0.06
Guatemala	2003 - 2021 (11)	0.15	Sweden	1974 - 2022 (41)	0.02
Guinea	2005 - 2010 (6)	0.08	Switzerland	1993 - 2020 (24)	0.08
Honduras	2003 - 2015 (10)	0.40	Tunisia	1997 - 2011 (12)	0.32
Hong Kong	1997 - 2011 (13)	0.04	Turkey	2011 - 2015 (5)	-0.16
Hungary	1997 - 2021 (21)	0.07	USA	1972 - 2021 (44)	0.02
Iceland	2002 - 2014 (12)	0.10	Ukraine	1997 - 2020 (23)	0.05
India	1997 - 2019 (16)	0.14	United Kingdom	1972 - 2021 (42)	0.02
Indonesia	2019 - 2019 (1)	-0.41	Uruguay	2017 - 2017 (1)	0.56
Iran	1998 - 2018 (19)	1.09	Uzbekistan	2012 - 2021 (9)	0.05
Ireland	1997 - 2021 (23)	0.02	Vanuatu	2003 - 2007 (3)	0.15
Israel	1997 - 2017 (18)	0.07	Venezuela	1999 - 2019 (20)	0.31

Note: θ is defined as $\frac{\Delta s^*}{\Delta g(K)}$. Thrift theory predicts $\theta \cong 1$. See Section 3 for predictions of free growth theory.

4. Inferences for the creative destruction idea

Creative destruction (Schumpeter (1912)[8]) is obsolescence of existing capital through innovation of new and preferred technology, as with horse-drawn carriages and automobiles. Granting the obsolescence of old ways by new, it does not follow that innovation reduces values of assets already present. Data cited in this paper appear to show that capital growth is maximized when investment is enough to offset depreciation. A natural source of that investment is depreciation of assets the new technology will replace, and a natural market for the new technology may be owners of the assets to be replaced. Carriage owners were better able to afford cars when the carriages in which they had invested before lived out their economic lives as originally expected, so that the depreciation component in the cash flows originally expected can pay for cars in replacement just as it might have paid for new carriages.

This analysis suggests that market forces favor a kind of orderly obsolescence through replacement, without premature displacement, as the new technology awaits its turn.

5. Solow's puzzle

Some growth is capital widening, where structures and implements increase in number but do not change in design. Capital widening, however, is practical only so far before glut and diminishing returns set in. Further growth must come from capital deepening, meaning improvements in the design of capital. Solow (1956) noted a kind of middle ground between capital widening and capital deepening in the disembodied growth mentioned earlier; ships carrying coals to Newcastle can raise prospective cash flows, and hence present value, by reversing the business plan. But Solow, who came to conclusions similar to ours from different evidence, puzzled as to how capital growth without net saving could be possible for capital deepening through "embodied" growth, where products of new design are made from plant of new design.

Solow received the Nobel Prize in 1987. His acceptance speech mentions the embodiment problem:[9]

"... much technological progress, maybe most of it, could find its way into actual production only with the use of new and different capital equipment. Therefore the effectiveness of innovation in increasing output would be paced by the rate of gross investment. A policy to increase investment would therefore lead to ... a faster transfer of new technology into actual production ... By 1958 I was able to produce a model that allowed for the embodiment effect. If common sense was right, the embodiment model should have fit the facts better than the earlier one. But it did not... I do not know if that finding should be described as a paradox, but it was at least a puzzle."

Solow was right in that much innovation requires months or years of negative cash flow, in the form of gross investment, in building new plant before the new products roll out from the factory doors. From arguments in Section 1, however, this investment would not have been forthcoming unless positive cash flows at the end of the tunnel were predicted to be sufficient to create positive present value from the start. Data in our displays, which show no effect of net saving (investment) on capital growth, suggest that this initial creation of value, plus depreciation investment, are enough.

6. Mechanics of free growth

First, in terms of Solow's puzzle, we suggest that embodied growth is disembodied growth on a finer scale. At each step toward realization of the new plant and products, raw materials and products and labor skills and plant capacity currently available on the market are adapted to new uses. The innovator pays for these inputs at a market price determined by their value in established productive uses, but applies them innovatively to realize higher prospective cash flows, and hence higher present values, to the innovator (Marshall (1890), Schumpeter and Opie (1934)). That new use of materials and skills applied could give the solution to Solow's puzzle with respect to practical feasibility, as in Section 1, while the economic advantage to the investor, here called innovator's reserve,[10] could give the solution with respect to source of payment.

Our findings support those of Piketty and Zucman (2014) and Kurz (2023) as to the market power of innovators to explain capital growth beyond net saving. Those studies assumed that net saving is recovered in capital growth, so

[8] Published by Schumpeter in 1912, and translated by Opie in 1934.

[9] Solow (1988). The terms capital deepening, capital widening, embodied growth and disembodied growth are all Solow's.

[10] i.e., capital and labor inputs are worth more to the innovator in that the innovator applies them in ways to realize greater returns. The present value of additional cash flow enabled by this advantage in return quantifies the innovator's reserve.

that this market power, which we call the innovator's reserve, need explain only the part of capital growth remaining. Again, we go farther by questioning the assumption that net saving contributes even a part of capital growth.

7. Tech Externalities

The innovator's reserve, quantified by the present value of the innovator's idea, gives the amount of free growth to the innovator. Others may benefit indirectly. Old factories and warehouses and residences gain in utility as they house higher-tech equipment and worker skills. Old highways do the same as they carry higher-tech freight and drivers and passengers to higher-tech destinations in higher-tech vehicles.[11]

Most externalities,[12] positive and negative, are part of random market noise expected to offset to zero over time and scale. The examples above, which we call tech externalities, can impose a prevailingly positive vector on capital value resisting the downward pull of depreciation. Free growth, in the full sense, means effects of the innovator's reserve on capital of the innovator, plus effects of tech externalities on capital in general.

8. Market value and book value

It is natural to reason that depreciation gives the only change in capital value, once investment has ended, apart from market noise converging to zero. Insofar as depreciation can be predicted, then, depreciated cost would show the norm (expectation) toward which current market value converges. It would follow that investors might anticipate the market by selling when prices are above depreciated cost, and buying when below. This reasoning would justify the reliance of modern national accounts, and the equations of macroeconomics (macro) which they were designed to inform, since the inception of both in Keyne's time, on depreciation cost as the preferred measure of capital value. [13]

We have argued that capital grows and declines with effects of the innovator's reserve and tech externalities, as well as investment and depreciation and market noise. Consider, however, a period when innovation is slight, so that depreciation and market noise give substantially the only changes in value of an investment once booked. It would follow that depreciated cost (of investment) could find the norm described, even if depreciation were predicted perfectly, only insofar as did original cost. An argument that optimal accounting can give a sounder measure of worth than market value would seem to need a supporting argument that price and worth were closer together at the time assets were acquired, and depreciation bases set at their cost, than over the depreciation period following. Garbage in, garbage out.

We do not question the value of depreciation accounting, or the importance of the depreciation concept in economic understanding. But we suggest that current market value gives a more reliable measure of capital, when both are known, than an assumption-dependent projection from past market value subject to the same criticisms as current market value.

These reflections help explain our own reliance on market value capital. They might also suggest rethinking of the logic by which macro and national account have preferred to reason replacement cost.

9. Optimum investment policy

Data and arguments suggest that the optimum amount of saving, at the global scale, is replacement saving to offset depreciation, and nothing more. That would not mean depreciation found by the depreciated cost method, which equates net investment to capital growth at all scales, and which cannot measure the contributions of the innovator's reserve or tech externalities. Up to a point, it should be possible to analyze the composition of market capital, and to model depreciation of the whole, just as national balance sheet accounting does with book value capital today. A better plan, as Solow (1956) wrote in response to Harrod's knife edge argument (Harrod, 1939), is to expect the market to maximize rate of return, and to sense the point where glut begins and returns fall.[14] Markets do so imperfectly when tax and other public policy reward saving over distributions and consumption. Findings in this paper suggest reconsideration of such policies. These include the double tax on dividends, and the greater tax rate on ordinary income than on capital gains. Effects of removing the double tax, and removing the difference between tax rates on ordinary income and on capital

[11] Rae (1834) makes essentially the same argument in terms of the technology of his time.

[12] Externalities are defined as effects of events on nonparticipants.

[13] As argued by James Tobin in 1968 and 1969. "Tobin's q" is the ratio of market to replacement value (RV). RV means depreciation cost under replacement cost accounting, where past investments and subsequent depreciation are re-expressed in the cheaper dollars of today. Investors were advised to invest when Tobin's q is less than one, and divest or withhold investment conversely.

[14] Harrod had argued that saving must hit the warranted rate exactly or risk positive feedback through the operation of the output/capital ratio (accelerator).

gains, could be revenue-neutral and non-partisan if the corporate tax were raised to match, if the tax rates on ordinary income and on capital gains met somewhere between, and if thoughtful grandfathering eased the transition.

10. Optimum government policy

Households dissave in consumption, while firms dissave through dividends and other distributions. Public policy, in many nations, has inhibited dissaving in the belief that such policies favor capital growth. Rather they inhibit consumption, add nothing to capital growth, and thus inhibit net output as a function of both together (see Appendix B).

The effect of repealing such policies would be to address capital glut not by investing less, but by disinvesting more.[15] This strategy spares government the burden of guiding investment, at least until the effects of the repeal are known.

11. Inferences for microeconomics

Microeconomics ("micro") studies the mutual effects of supply, demand and price. It teaches that producer cost, including interest as the cost of time, tends to equilibrate to market value realized. We have proposed an exception in the innovator's reserve; the innovator pays the going rates for skills and materials needed, but applies them to make products of greater market value.

12. Inferences for inflation

If free growth theory is correct, then net investment adds a superfluous cost to capital growth, while decreasing market value per unit through capital glut. Both effects would tend to contribute directly to capital inflation, and indirectly to inflation of consumer goods by adding depreciation costs. This consideration adds to the case for urgent attention to our argument, and for more study to confirm or refute it.

13. Net output

In Keynesian tradition, net output equals consumption plus net investment. We have disputed this reasoning in our argument that net investment adds nothing to capital growth, inhibits consumption, and thus tends to reduce the sum of both. We also critique it on another ground. Assume, for the sake of argument, that we are wrong, and that net investment realizes equal capital growth. Net output means value added. Appendix B will show an argument that to interpret value added as the sum of consumption and capital growth, and neither less nor more, would miss two components, one negative (human depreciation) and one positive (self-invested work), in value added to human capital. Both of these flows are unmeasurable by known means. Thus we have not sought to measure net output or its growth, and have treated capital growth alone.

14. Data sources

All our data are drawn from the free online website World Inequality Database (WID). This source collates data from national accounts and tax data of 105 countries in constant currency units, and adjusts them where needed to conform to current standards of the System of National Accounts (SNA) published by the United Nations. We show results for the 92 of those countries which report both factors, net saving and market-value capital, needed for deriving the thrift indexes. WID's source for these data is national accounts.

Net saving S_{net} and market-value K are taken from net national saving (msavin) and market-value Capital Wealth (mnweal) respectively. GDP, which we use only for weighting purposes in Figs. 1 and 2, is reproduced from GDP (mgdpro).

15. Accessing our results and methods

Tables and other displays of our findings for each country, and showing our methods of calculation, can be accessed at the web appendix (https://web-appendix.shinyapps.io/Sources_of_Capital_Growth).

[15] Higher dividends and distributions by firms, that is, would enable higher consumption by households.

16. Displays

The web appendix includes displays of θ for each year in each country over the report period. These tend to show upward and downward spikes in values of θ in some years. Those spikes tend to be associated with small absolute values of denominators, in these cases $\Delta g(K)$, in those countries and years. Small denominators magnify errors in measurements of numerators. Worse, when denominators are small, small mismeasurements in them might reverse the term in sign.

To maximize reliability of test results, we apply a range of screens to omit years where absolute denominators fall below a given threshold. Some displays in the web appendix show data for all years, regardless of denominator size. Others screen out all years where absolute denominators are less then 0.01, then 0.025, then continuing upward in increments of 0.025 to a maximum screen of 0.15. θ is plotted for each country unscreened and at each of the seven successive levels of screening. All displays in the main body of this paper apply a screen of 0.01. The denominator whose absolute value is screened is capital acceleration $\Delta g(K)$ or capital growth rate $g(K)$ in all displays.

Screening out years where absolute $\Delta g(K)$ or $g(K)$ is small would cost little in informative value even if measurements were exact. In those years, there is little capital acceleration or capital growth, positive or negative, for either thrift theory or free growth theory to explain. Market noise alone might account for $\Delta g(K)$ or $g(K)$ in such years. Screening reduces the number of observations, but increases the reliability and informative value of each.

17. Disclaimers

We accept that capital growth is impossible without capital replacement first, to make up losses to depreciation, and that the expected source of capital replacement is replacement investment. Thus we accept the necessity and efficacy of replacement investment. We dispute only the efficacy of net investment in realizing capital growth after replacement investment has made up for depreciation.

Evidence supports the proposal of Modigliani (1954) that the young save to afford families and retirement, and that retirees dissave to pay for both as they arrive.[16] Such age-based differences are expected to offset in the stationary state, where tradition and free growth theory agree that overall net saving is zero. We accept that individual net saving can realize equal individual capital growth, so long as collective saving does not exceed collective dissaving.

We do not assert that net saving and capital growth are uncorrelated. We suggest that they are causally related in that negative net saving, meaning a deficit in depreciation investment, imposes an equal downward vector on capital growth. This causal relatedness appears in the thrift index and the regression of s^* on $g(K)$ shown in our displays. Rather, we conclude that positive net saving has no effect on positive capital growth.

18. Discussion and conclusions

We have argued that the innovator's reserve and tech externalities pay for capital growth as a whole, and that net saving by some can realize equal capital growth only insofar as others dissave as much. We adopted the Modigliani model, where young households save to afford families and retirement while older households dissave to pay for both as they arrive, as an example of this saving/dissaving balance. We cannot prove these hypotheses directly, but offer them as possible explanations of evidence presented here that net saving at the overall scale adds nothing to capital value.

These findings challenge the teachings that capital growth at all scales is effected by net saving enabled by consumption restraint, and that producer cost, including imputed interest as the opportunity cost of capital, converges to market realization. Evidence suggests that overall capital growth is free, and consequently that market realization, in the presence of innovation, exceeds producer cost by the entirety of capital growth. Meanwhile the same evidence, which indicates that net saving adds no capital value, suggests a review of the teachings that consumption plus net saving gives net income, and that consumption plus net investment gives net output. Appendix B will also question the latter teachings.

Embodied growth is disembodied growth on a finer scale. It redeploys or repurposes existing labor skills, raw materials, and plant capacity, as well as existing finished goods, to achieve higher returns than available from the customary uses which determine their prices. The present value of yields from this advantage in return, or equivalently the innovator's reserve, plus tech externalities, defines the non-random component in free growth.

[16] Friedman (1957) follows a different argument to a similar conclusion.

Solow's puzzle runs deep. How can thoughts be things? How can assessments of cash flows over the future pay for their realization in the present, even when plant of new design must come first? Science and philosophy push our minds into areas where they are not at home. Neither relativity nor quantum mechanics, which explore the scale of great speeds and minute distances, sink in easily. Economics, too, invites surprises as it looks farther from the scale of the household and community and firm to which we are accustomed. Free growth theory predicts only at the universal scale, and accepts that thrift explains a greater part of growth at progressively smaller scales. Our findings, with that allowance, are awkward for much of economics as generally taught. We encourage others to check our data and methods and inferences, and improve on them wherever they can.

Appendix A. Equivalence of saving and investment

Keynes (1936) argued correctly that saving, if not invested in currently produced producer goods and services, adds nothing to capital, and thus nothing to output. He inferred that uninvested saving has no value, as it accomplishes no growth, and described it as "intended saving." True saving, by that reasoning, equalled saving actually invested.

We suggest that investors seek to maximize return, within risk tolerance, and will sometimes hold saving in cash or equivalents at zero return in recessions and depressions when positive returns cannot be found, and so when investing would tend to reduce capital and output rather than increase them. Thus we do not share the Keynesian view that saving need be invested in order to maximize its value. Saving will tend to equal investment in normal times, when prospective returns bring animal spirits, and could include saving in cash when not. It is only for these profitable times that we accept the equality of saving and investment, and only as a simplifying assumption.[17]

Appendix B. Net output with human capital

Human capital is impractical to measure, as it leaves little market record other than for its rental income in pay and investment cost in nurture and schooling. Thus national accounts leave it implicit, and allow us to infer what we can from data for pay and nurture and schooling. Those accounts are founded on the principle, sound when terms are appropriately redefined, that net output, or value added, is expressed in the sum of capital growth and net outflow from the value-added chain. In national accounts, then, where physical capital is the whole of capital while net outflow of the chain is the whole of consumption, the reasoning is

$$Y = \Delta K + C\,, \quad \text{neglecting human capital.} \tag{B.1}$$

It is possible in principle to model a value-added chain which includes human capital, and to compare findings with those shown in Eq. (B.1). Let human capital H, in that new model, stand as the last link in the value-added chain. Adapting the classic illustration of the value added principle, say that farms produce wheat, mills convert the wheat to flour, bakeries convert the flour into bread, and humans convert some of the bread, called invested consumption, into human capital. The net outflow from this extended value-added chain is not all of consumption, but only pure consumption, meaning the part remaining after the part invested in human capital (invested consumption) is subtracted.[18] By this reasoning, the principle that net output is expressed in capital growth plus net outflow gives

$$Y = \Delta K + \Delta H + C_p\,, \quad \text{allowing human capital,} \tag{B.2}$$

where C_p gives pure consumption.

Yoram Ben-Porath (1967) reasoned that growth in human capital equals invested consumption plus self-invested work less human depreciation[19]. Let C_s, W_s and $D(H)$ show these flows respectively. Thus the combined arguments of Schultz and Ben-Porath arrive at

$$C = C_s + C_p \quad \text{and} \quad \Delta H = C_s + W_s - D(H)\,, \quad \text{allowing human capital.} \tag{B.3}$$

[17] The weakness in the Keynes argument, if we are right, was his inference that net saving finds no utility, and hence no economic value, except as invested to realize capital growth. More exactly, we propose the utility of net saving is to maximize capital growth according to investment opportunities, and to maximize it by withholding investment when all investments promise loss.

[18] The concepts of invested consumption, pure consumption, self-invested work and human depreciation were introduced in Schultz (1961).

[19] Equation 4 in Ben-Porath's paper, summarizing his first three equations. His terms and notation differ from ours. He expressed his idea in Cobb-Douglas functions, whose values vary with scalars and exponents chosen, as opposed to our simpler algebra. We believe, however, that meanings agree.

Substitution of these equations into Eq. (B.2) finds

$$Y = \Delta K + C_s + W_s - D(H) + C_p \quad \text{and consequently}$$

$$Y = \Delta K + C + W_s - D(H)\,, \quad \text{allowing human capital,} \tag{B.4}$$

if Schultz and Ben-Porath and the reasoning here are right.

Appendix C. Mill's statement of the free growth idea

Mill (1848), book 1, chapter 5, section 4, includes:

> If it were said, for instance, that the only way to accelerate the increase of capital is by increase of saving, the idea would probably be suggested of greater abstinence, and increased privation. But it is obvious that whatever increases the productive power of labour creates an additional fund to make savings from, and enables capital to be enlarged not only without additional privation, but concurrently with an increase of personal consumption.

This passage may be the first clear statement of what we call free growth theory. Mill's use of the words "accelerate" and "concurrently" suggest that his path of reasoning was something like ours, and that he meant what we call free growth rather than the alternating phases of higher and lower saving rates pictured in thrift theory.

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